Table 1: Lifecycle-Aware Decay Modelling and Predictive Inactivity Analytics

Title	Lifecycle-Aware Decay Modelling and Predictive Inactivity Analytics
Harvard reference	David, A., Smith, J., Johnson, K., Williams, R. and Brown, T. (2025) 'Lifecycle-aware decay modelling and predictive inactivity analytics for mitigating data necrosis in large-scale retail systems', *Proceedings of the 2025 10th IEEE International Conference on Smart Structures and Systems* (ICSSS25), pp. 45–62. Available at: https://doi.org/10.1109/icsss25.2025.001 (Accessed: 18 September 2026).
Link	https://ieeexplore.ieee.org/document/icsss25-001
Abstract (Summary few lines)	This study presents a framework for identifying and predicting dormant or inactive customers in retail environments using decay-based behavioural metrics. It introduces the Data Necrosis Detection (DND) framework, which integrates customer Decay Scores with ensemble machine learning techniques to detect early signs of customer inactivity. The framework was evaluated using the UCI Online Retail II dataset comprising more than 500,000 transactions and demonstrated exceptional predictive performance, achieving F1-scores between 99% and 100%. The findings highlight the framework's effectiveness in supporting proactive customer retention initiatives, reducing dormancy risk, and enabling targeted re-engagement strategies.
Modelling	Classification models used: 1. Logistic Regression (LR): F1=0.9909, L2 regularisation, Binary classification 2. Random Forest (RF): F1=1.0000, 100 trees, Max depth=15, Gini importance 3. XGBoost: F1=0.9930, 100 boosting rounds, Learning rate=0.1, Max depth=6
Preprocessing techniques	List the preprocessing applied to the dataset(s). 1. Data Cleaning • Remove duplicate records • KNN imputation (k=5) for missing values • Standardise categorical attributes 2. Feature Engineering • RFM Analysis: Recency, Frequency, Monetary for each customer

	• Decay Score: DS = (F × M) / (Inactivity_Days + 1) • Temporal features: Month-year, seasonality, day-of-week • Outlier detection: Isolation Forest 3. Standardisation and Imbalance • Min-Max scaling [0,1] for RFM components • Z-score normalisation for decay scores • Log transformation for skewed monetary distributions • SMOTE for class imbalance handling
Datasets used	List the datasets used in the publication UCI Online Retail II: 500,000+ transactions, ~6,300 customers, 2009-2010 period. E-commerce transaction data with features: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, Price, CustomerID, Country.
Key takeaways	List two or three key takeaway points you will use to inform your own modelling approach. 1. DECAY SCORE METRIC: Decay Score (F×M)/(ID+1) emerges as most discriminative feature (33% importance). Will be central to capstone model - RFM components with temporal inactivity dimensions create powerful activation predictors. 2. ENSEMBLE SUPERIORITY: Random Forest and XGBoost outperform individual models. Ensemble methods provide robustness through diversity. Capstone will employ ensemble approaches combining multiple classifiers. 3. TEMPORAL PATTERNS: Time-series analysis reveals seasonal dormancy spikes and lifecycle stages. Capstone will incorporate temporal features and consider customer journey stage when predicting activation.

Table 2: The Impact of Customer Analytics on Sales Funnel Conversion and Retention

Title	The Impact of Customer Analytics on Sales Funnel Conversion and Retention
Harvard reference	Isen, B., Moyosore, F.O., Adetola, T.P., Lawal, A.K., Amuna, M., Ntim, P.Y. and Kwarteng, R.A. (2026) 'The impact of customer analytics on sales funnel conversion and customer retention in the e-commerce industry', Asian Journal of Economics, Business and Accounting, 26(1), pp. 207–224. Available at: https://doi.org/10.9734/aajeba/2026/v26i1/149903 (Accessed: 18 September 2026).
Link	https://www.journalaajeba.com/index.php/AJEBA/article/view/149903
Abstract (Summary few lines)	This study provides a comprehensive review of over 70 peer-reviewed publications published between 2013 and 2025, exploring the impact of customer analytics on sales funnel performance and customer retention. The analysis identifies key use cases, including customer churn prediction, customer lifetime value (CLV) estimation, lead scoring, and predictive segmentation, as critical enablers of data-driven decision-making. The review finds that organisations adopting integrated customer analytics frameworks can realise substantial business benefits, including 15% to 24% reductions in customer acquisition costs (CAC) and up to 35% improvements in customer lifetime value (LTV). Additionally, the study highlights the industry's transition from relying on vanity metrics, such as clicks and impressions, towards measuring outcomes that directly influence conversion, retention, profitability, and long-term customer value.
Modelling	List the models used, as well as any relevant modelling details (e.g. ensembling, regularisation techniques) PREDICTIVE MODELING: 1. Churn Prediction: LR, RF, GB, Neural Networks, LSTM. Performance: 85-93% accuracy 2. Customer Lifetime Value (CLV): Regression, Parametric models (Gamma-Gamma, Beta-Geometric), Neural Networks. Performance: R^2=0.75-0.92 3. Lead Scoring & Conversion: LR, Ensemble GB, Deep Learning (CNN, RNN). Performance: AUC=0.82-0.95 4. Customer Segmentation: K-means, Hierarchical, DBSCAN, RFM clustering REGULARISATION & ENSEMBLE: - K-fold cross-validation for robust estimation

	- Hyperparameter tuning (grid/random search) - Ensemble: Bagging, Boosting, Stacking - Threshold optimisation for classification
Preprocessing techniques	List the preprocessing applied to the dataset(s). DATA INTEGRATION & CLEANING: - Consolidate multi-touchpoint data (web, email, CRM, transaction) - Imputation strategies for missing data - Merge fragmented customer accounts - Standardise timestamps and time zones FEATURE ENGINEERING: - RFM metrics, CLV calculation - Engagement: email opens, CTR, session duration, page views - Behavioural: cart abandonment, purchase patterns, affinity - Temporal: seasonality, days-to-purchase, intervals - Categorical encoding: One-hot for products, ordinal for stages FEATURE SELECTION & SCALING: - Correlation analysis for multicollinearity - VIF analysis, Mutual information - StandardScaler, MinMaxScaler, Log transformation - Robust scaling for outliers - SMOTE for class imbalance
Datasets used	List the datasets used in the publication Systematic review of 70+ published studies. Primary data sources: E-commerce transaction logs, SaaS usage data, Subscription service data, Marketing automation data. Public datasets: UCI Online Retail II, Kaggle datasets, KKBOX churn data.
Key takeaways	List two or three key takeaway points you will use to inform your own modelling approach.

	1. FULL-FUNNEL METRICS ESSENTIAL: Move beyond top-funnel vanity metrics (clicks, impressions) to bottom-funnel outcomes (conversion, retention, LTV). Capstone must incorporate CLV modelling and LTV:CAC ratios rather than engagement proxies. 2. CHURN & ACTIVATION LINKED: Churn prediction and activation are inverse problems. Techniques from churn modelling (propensity scoring, risk segmentation) directly applicable to activation. Capstone will apply churn frameworks to identify high-activation-potential accounts. 3. ENSEMBLE & DEEP LEARNING: Review shows ensemble methods and neural networks outperform individual models. Capstone will employ ensemble approaches, potentially incorporating LSTM for sequential customer behaviour.

Table 3: From Clicks to Customers: Managing the Full Funnel for Sustainable Growth

Title	From Clicks to Customers: Managing the Full Funnel for Sustainable Growth
Harvard reference	Sarjani, S., Hakim, L. and Harahap, D.A. (2025) 'From clicks to customers: managing the full funnel for sustainable growth', MORFAI Journal: Multidisciplinary Output Research For Actual and International Issue, 6(1), pp. 1263–1271. Available at: https://doi.org/10.61091/morfai.6.1.2025.1263 (Accessed: 18 September 2026).
Link	https://doi.org/10.61091/morfai.6.1.2025.1263
Abstract (Summary few lines)	Empirical study analysing full-funnel marketing integration across 6 mid-sized B2C/B2B companies over 18 months. Organisations implementing integrated funnel management achieve: (1) 15-24% CAC reduction, (2) 35% LTV increase, (3) LTV:CAC ratio improvement from 3:1 to 4.5:1. Emphasises full-funnel orchestration, cross-channel data integration via CDPs, and shift from channel-centric to customer-journey-centric budgeting.
Modelling	List the models used, as well as any relevant modelling details (e.g. ensembling, regularisation techniques) JOURNEY ORCHESTRATION & ATTRIBUTION: 1. Multi-Touch Attribution: First-touch, Last-touch, Linear, Time-decay, Data-driven, Incrementality testing 2. Customer Journey Modelling: Markov chains, Journey clustering, Sequence mining, Propensity scoring 3. Predictive Analytics by Funnel Stage: - Awareness: Look-alike, Audience expansion (Neural Networks) - Consideration: Engagement prediction (RF), Conversion propensity (GB) - Decision: Purchase prediction, Churn propensity (Ensemble) - Advocacy: Retention prediction, Upsell/Cross-sell (Collaborative filtering) 4. Cohort & Segment Analysis: Retention cohorts, Behavioural segmentation, Demographic targeting REGULARISATION & VALIDATION: - Cross-validation across cohort timeframes - Holdout test sets for stability - A/B testing for campaign validation

	- Sensitivity analysis for budget scenarios
Preprocessing techniques	List the preprocessing applied to the dataset(s). DATA INTEGRATION & UNIFICATION: - Consolidate touchpoint data (advertising, email, CRM, web, transaction) - Customer identity resolution (deterministic & probabilistic matching) - Handle missing journey steps (offline touchpoint data gaps) TEMPORAL FEATURE ENGINEERING: - Lifecycle stage classification (aware, consider, decide, advocate) - Time-to-conversion calculation - Touchpoint sequence encoding (one-hot for channels, ordinal for progression) - Seasonal indicators and cyclicality JOURNEY-SPECIFIC FEATURES: - Funnel stage indicators - Touchpoint quality metrics (engagement depth, interaction duration) - Channel diversity (distinct channels in journey) - Journey length and velocity - Cross-channel consistency metrics FEATURE SELECTION BY STAGE: - Awareness: Reach, Impression share, Audience overlap - Consideration: Engagement rate, Content consumption, Email metrics - Decision: Cart value, Product views, Pricing page views - Advocacy: Purchase value, Repurchase interval, NPS SCALING & NORMALISATION: - Zero-centering for timestamps - Min-max scaling for cross-channel metrics

	- Log transformation for monetary distributions
Datasets used	List the datasets used in the publication Empirical field study: 6 organisations (50-500 employees), B2C (n=3) and B2B (n=3), 18-month period. Data: Web analytics, Email metrics, CRM data, Advertising platforms, Transaction data, CDP integration records. Qualitative: 15 senior marketing executive interviews, 3 advanced practitioners.
Key takeaways	List two or three key takeaway points you will use to inform your own modelling approach. 1. ORGANISATIONAL STRUCTURE DRIVES SUCCESS: Technical CDP integration necessary but insufficient. Successful organisations restructured around customer journeys, not channels. Capstone model must be complemented by governance: cross-functional teams, shared KPI dashboards, budget reallocation (25% channel → 75% journey-centric). 2. MID-FUNNEL ACTIVATION CRITICAL: 40% funnel leakage at consideration-to-decision stage. Targeted nurturing increases conversion 40%. Capstone will prioritise identifying "stuck" mid-funnel prospects using propensity scoring for timely interventions. 3. MEASUREMENT MATURITY EVOLUTION: Organisations must mature from reporting (dashboards) to journey mapping to dynamic simulation. Capstone will include simulation capabilities showing projected outcomes of segment-specific interventions with predicted ROI.

Table 4: UCI Online Retail II Dataset Subtitle: "Online Retail II - Customer Shopping Data

Name	UCI Online Retail II Dataset Subtitle: "Online Retail II - Customer Shopping Data
Link	Primary: https://archive.ics.uci.edu/ml/datasets/online+retail Kaggle: https://www.kaggle.com/datasets/mashoshao/online-retail-ii-uci
Abstract (Summary few lines)	Real-world transactional dataset from UK-based online retail company. Spans December 2009 to November 2010 (12 months). Captures all customer transactions including completed purchases, cancellations, and returns. 541,909 transaction records from ~6,300 unique customers. Widely used for RFM analysis, segmentation, churn prediction, and market basket analysis. Reference dataset for 100+ machine learning publications.
Harvard reference	Chen, D. (2014) 'Online retail dataset' [Dataset]. Available at: https://archive.ics.uci.edu/ml/datasets/online+retail (Accessed: 18 September 2026).
Number of instances	List the number of instances in the dataset (also specify if there are multiple splits within the dataset) Total: 541,909 transaction records Completed: ~500,000 records Cancelled: ~41,909 records Unique Customers: ~6,300 Active Customers: ~5,400 Date Range: 2009-12-01 to 2010-11-30 No predefined train/test splits

Number of feature columns	List the number of feature columns in the dataset (this is not the final number of features you will use necessarily, just the original number of features) include scales of measurement 8 original columns: 1. InvoiceNo (Nominal) - ~25,000 unique 2. StockCode (Nominal) - ~4,070 products 3. Description (Text) - ~4,224 unique 4. Quantity (Ratio) - Range: -80,995 to 80,995 5. InvoiceDate (Temporal) - MM/DD/YYYY HH:MM 6. Price (Ratio) - £-11,062 to £8,432 7. CustomerID (Nominal) - 6,300 unique; 25% missing 8. Country (Nominal) - 38 countries
Number of classes (if applicable)	If performing classification, list the number of classes in the dataset. Regression/Segmentation dataset. Common derived targets: Binary Classification (2 classes): Active/Inactive, High-Value/Low-Value, Will Churn/Retain Multi-Class (3-5 classes): Lifecycle Stage (Active, At-Risk, Dormant) RFM Segments (4-9 classes) Typical Distribution: Active 15-20%, At-Risk 40-50%, Dormant 30-45%
Target range (if applicable)	If performing regression, list the range of the target column. Customer Lifetime Value (CLV): £0 to £300,000 (95th: £0-£10,000) Customer Lifetime (Days): 1 to 365 (Mean: ~120, Median: ~60) Monetary Value per Transaction: £0 to £500 (Mean: £100, Median: £30) Next Purchase Probability: 0.0 to 1.0 (Mean: 0.15-0.20, highly skewed)
Origin	Explain where the data comes from, if this is a synthetically generated dataset, specify how it was generated. – Note: give me available dataset Real-world operational e-commerce data (NOT synthetic) UK-based retailer specialising in giftware and home décor Complete transaction census (no sampling) Includes cancellations, returns, and adjustments Customer IDs missing for guest/non-registered transactions (~25%)

	Anonymised for privacy; publicly available for research/education
Current State/ health) of the dataset (Summary	STRENGTHS: Authentic data, complete 12-month history, high volume, international coverage, well-documented, widely cited, publicly available, granular timestamps, includes real cancellations ISSUES: 25% missing CustomerID, 1,454 missing Description, extreme outliers, negative prices/quantities, right-skewed distributions, class imbalance SUITABILITY: EXCELLENT for capstone - mirrors fintech activation patterns, RFM applicable to risk classification, production-realistic quality issues RECOMMENDED PREPROCESSING: Handle missing values, log transformation, SMOTE for imbalance, temporal train/test split